

2.3 Trigonometric Substitutions

Motivating Example:

Suppose we wish to find $\int \sqrt{a^2 - x^2} \, dx$ where a is some positive real number.

One question we can ask about this integral is: Is there a substitution we can make that will simplify the integration process?

Table of Trigonometric Substitutions:

Expression	Substitution	Identity

Example 1 (you try): Suppose $x = a \sec(\theta)$ where $0 \leq \theta < \frac{\pi}{2}$ or $\pi \leq \theta < \frac{3\pi}{2}$ and a is positive. Write $\sqrt{x^2 - a^2}$ in terms of θ (fully simplify your answer).

Example 2: Find $\int \frac{\sqrt{9-x^2}}{x^2} dx$.

Example 3: Find $\int \frac{dx}{\sqrt{x^2 - 4}}$

Example 4: $4 \int_{-\frac{1}{2}}^{\frac{1}{2}} \sqrt{\frac{1}{4} - x^2} \, dx$

Example 5 (you try):

(a) Find the exact value of: $\int_0^1 \frac{dx}{(x^2+1)^2}$. *Hint:* $\int_0^1 \frac{dx}{(x^2+1)^2} = \int_0^1 \frac{dx}{(\sqrt{x^2+1})^4}$

(b) In no more than three sentence explain how you would find $\int \frac{x}{\sqrt{1+x^2}} dx$.

Example 6:

Consider $\int_{-3}^1 \frac{x}{\sqrt{3-2x-x^2}} dx$. Complete the square and perform a substitution to set up (do not evaluate) an equivalent integral to which trigonometric substitution can be applied.